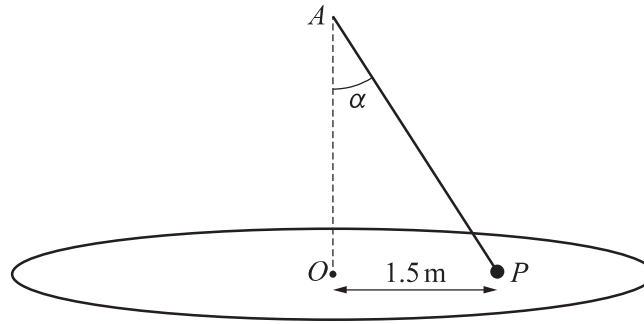




A rough horizontal disc, centre O , rotates with constant angular speed $\omega \text{ rad s}^{-1}$. A particle P of mass 1.6 kg lies on the disc at a distance 1.5 m from O , and is attached to a point A vertically above O by a light elastic string. The string has natural length 2 m , modulus of elasticity 32 N and makes an angle α with the vertical OA (see diagram). Particle P moves in a horizontal circle also at a constant angular speed $\omega \text{ rad s}^{-1}$. Particle P is on the point of slipping in the direction OP . The coefficient of friction between the particle and the disc is 0.5 .

- (a)** Given that the tension in the string is 8 N, show that $\sin \alpha = 0.6$. [2]

[illegible]



[5]

[illegible]